

CURIONEST

Acids & Bases: Buffers, Titration & Salt Hydrolysis

Complete Lesson Pack · Henderson–Hasselbalch · Equivalence Points · Salt Ions

WHAT'S INSIDE

- ✓ Complete concept notes with worked examples
- ✓ Practice sets in varied formats (short answer, MCQ, calculations)
- ✓ Key terms glossary + full answer key with worked solutions
- ✓ Original figures and Word-native equations — print-ready & editable

Made by a chemistry teacher — aligned to the standard US chemistry sequence.

Learning Objectives

1. Explain how a buffer resists pH change and calculate buffer pH using the Henderson–Hasselbalch equation
2. Describe buffer capacity and what happens when acid or base exceeds buffer capacity
3. Predict whether aqueous solutions of salts are acidic, basic, or neutral
4. Interpret acid–base titration curves and locate the equivalence point and half-equivalence point
5. Choose an appropriate indicator for a given titration

Concept 1 — Buffers and the Henderson–Hasselbalch Equation

A buffer is a solution containing appreciable amounts of a weak conjugate acid–base pair, such as CH_3COOH and CH_3COO^- . It resists large pH changes when small amounts of acid or base are added.

The Henderson–Hasselbalch equation gives the pH of a buffer: $\text{pH} = \text{pK}_a + \log([\text{A}^-]/[\text{HA}])$. When $[\text{A}^-] = [\text{HA}]$, $\text{pH} = \text{pK}_a$.

Buffer capacity is the amount of acid or base a buffer can neutralize before the pH changes significantly. Capacity is greatest when $[\text{HA}]$ and $[\text{A}^-]$ are both large and roughly equal.

Example buffer: a solution of acetic acid and sodium acetate ($\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$). The initial pH of an equimolar buffer equals pK_a of acetic acid, 4.74.

★ Worked Example

A buffer contains 0.50 M CH_3COOH and 0.50 M CH_3COONa . $K_a = 1.8 \times 10^{-5}$. (a) What is the buffer pH? (b) What is the pH after 0.020 mol of HCl is added to 1.0 L of buffer?

Solution: (a) $\text{pK}_a = -\log(1.8 \times 10^{-5}) = 4.74$. $\text{pH} = 4.74 + \log(0.50/0.50) = 4.74$. (b) Added H^+ converts CH_3COO^- to CH_3COOH : $[\text{CH}_3\text{COO}^-] = 0.48 \text{ M}$, $[\text{CH}_3\text{COOH}] = 0.52 \text{ M}$. $\text{pH} = 4.74 + \log(0.48/0.52) = 4.71$. pH changed by only 0.03 units — the buffer worked.

★ **Common mistake:** Using the Henderson–Hasselbalch equation for strong acid/base solutions (it only applies to buffers of weak conjugates).

★ **Common mistake:** Forgetting to use the NEW concentrations after acid/base is added — always do the stoichiometry first.

Concept 2 — Hydrolysis of Salts: Acidic, Basic, or Neutral?

Salt ions can react with water (hydrolyze) and change solution pH. The conjugate of a weak acid (e.g., CH_3COO^- , CN^-) is a weak base; the conjugate of a weak base (e.g., NH_4^+) is a weak acid.

Rules: (1) A cation that is the conjugate acid of a weak base $\rightarrow$ acidic (e.g., NH_4^+ in NH_4Cl). (2) An anion that is the conjugate base of a weak acid $\rightarrow$ basic (e.g., CN^- in KCN). (3) Ions of strong acids/bases (Cl^- , Na^+) are neutral spectators. (4) Small, highly charged metal cations (Al^{3+} , Fe^{3+}) make solutions acidic by binding water and releasing H^+ .

Example: $\text{NH}_4\text{Br} \rightarrow \text{NH}_4^+$ (weak acid, from NH_3) + Br^- (neutral, from HBr) $\rightarrow$ the solution is acidic. $\text{KNO}_2 \rightarrow \text{K}^+$ (neutral) + NO_2^- (weak base, from HNO_2) $\rightarrow$ the solution is basic.

★ Worked Example

Predict whether aqueous solutions of the following salts are acidic, basic, or neutral: (a) $\text{Al}(\text{NO}_3)_3$ (b) RbI (c) KHCO_2 (d) $\text{CH}_3\text{NH}_3\text{Br}$

Solution: (a) Al^{3+} hydrolyzes $\rightarrow$ acidic. (b) Rb^+ and I^- are neutral spectators $\rightarrow$ neutral. (c) HCO_2^- is the conjugate base of a weak acid $\rightarrow$ basic. (d) CH_3NH_3^+ is the conjugate acid of a weak base $\rightarrow$ acidic.

★ **Common mistake:** Assuming every salt is neutral — check each ion's parent acid/base.

★ **Common mistake:** Treating metal cations like Na^+ the same as Al^{3+} (small highly charged cations are acidic).

Concept 3 — Acid–Base Titrations and Titration Curves

A titration curve plots pH versus volume of added titrant. For a strong acid + strong base, pH jumps sharply near the equivalence point ($\text{pH} \approx 7$). For a weak acid + strong base, the equivalence point is above pH 7; for a weak base + strong acid, it is below pH 7.

At the half-equivalence point, half the analyte has been neutralized: $[\text{HA}] = [\text{A}^-]$ and $\text{pH} = \text{pK}_a$. This is the flattest part of the curve (maximum buffering).

Indicators are weak acids whose color depends on pH. Choose an indicator whose color-change range ($\text{pK}_a \pm 1$) brackets the equivalence-point pH.

★ Worked Example

In the titration of 40 mL of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) with 0.100 M NaOH , predict the pH at: (a) 0 mL added (b) 20 mL added (c) 40 mL added.

Solution: (a) Initial weak acid: $[\text{H}_3\text{O}^+] \approx \sqrt{K_a \cdot C_a} = \sqrt{(1.8 \times 10^{-5} \times 0.100)} = 1.3 \times 10^{-3} \text{ M}$, $\text{pH} = 2.87$. (b) At 20 mL, half-equivalence: $\text{pH} = \text{p}K_a = 4.74$. (c) At 40 mL, equivalence: all acetic acid converted to CH_3COO^- (0.050 M). $K_b = K_w/K_a = 5.6 \times 10^{-10}$; $[\text{OH}^-] = \sqrt{K_b \cdot C_b} = \sqrt{(5.6 \times 10^{-10} \times 0.050)} = 5.3 \times 10^{-6} \text{ M}$; $\text{pOH} = 5.28$, $\text{pH} = 8.72$.

★ **Common mistake:** Assuming the equivalence-point pH is always 7 (true only for strong acid + strong base).

★ **Common mistake:** Using initial acid concentration instead of the diluted concentration after titrant is added.

Part A — Buffer pH Calculations (Questions 1–6)

1. A buffer is prepared from 0.200 M CH_3COOH and 0.200 M CH_3COONa ($K_a = 1.8 \times 10^{-5}$). What is its pH?
2. A buffer is prepared from equal volumes of 0.200 M CH_3COOH and 0.600 M CH_3COONa . (a) What is the pH? (b) Is the solution acidic or basic?
3. What is the pH of a buffer containing 0.250 M NH_3 and 0.350 M NH_4Cl ? $K_b(\text{NH}_3) = 1.8 \times 10^{-5}$.
4. How much solid sodium acetate must be added to 0.300 L of 0.50 M CH_3COOH to make a buffer with $\text{pH} = 5.00$? (Assume negligible volume change.)
5. What mass of NH_4Cl must be added to 0.750 L of 0.100 M NH_3 to give a buffer with $\text{pH} = 9.26$? (Assume negligible volume change.)
6. A buffer contains 0.40 M HF and 0.40 M NaF ($K_a = 6.8 \times 10^{-4}$). What is the pH after 0.010 mol of NaOH is added to 1.0 L of buffer?

Part B — Salt Hydrolysis (Questions 7–11)

7. Determine whether aqueous solutions of these salts are acidic, basic, or neutral: (a) FeCl_3 (b) K_2CO_3 (c) NH_4Br (d) KClO_4
8. Determine whether aqueous solutions of these salts are acidic, basic, or neutral: (a) $\text{Al}(\text{NO}_3)_3$ (b) RbI (c) KHCO_2 (d) $\text{CH}_3\text{NH}_3\text{Br}$
9. Write the hydrolysis equation for the acetate ion, CH_3COO^- , in water and identify the solution as acidic or basic.
10. Explain why the neutralization of a strong acid with a weak base gives a weakly acidic solution. Use $\text{NH}_3 + \text{HCl}$ as your example.

11. Explain why the neutralization of a weak acid with a strong base gives a weakly basic solution. Use $\text{CH}_3\text{COOH} + \text{NaOH}$ as your example.

Part C — Titrations (Questions 12–16)

12. Explain how to choose an appropriate indicator for the titration of a weak base with a strong acid.

13. Explain why an indicator changes color over a range of pH values rather than at a single pH.

14. For the titration of 40 mL of 0.100 M CH_3COOH ($K_a = 1.8 \times 10^{-5}$) with 0.100 M NaOH, calculate the pH after 20 mL of NaOH has been added.

15. For the same titration (40 mL of 0.100 M CH_3COOH with 0.100 M NaOH), calculate the pH after 40 mL of NaOH has been added.

16. For the titration of 40 mL of 0.100 M CH_3COOH with 0.100 M NaOH, calculate the pH after 41 mL of NaOH has been added (excess strong base).

Key Terms

Term	Definition
buffer	A solution containing a weak conjugate acid–base pair that resists pH change
Henderson–Hasselbalch equation	$\text{pH} = \text{pK}_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$ for buffer solutions
buffer capacity	The amount of acid or base a buffer can neutralize before pH changes significantly
hydrolysis	Reaction of a salt ion with water to produce H_3O^+ or OH^-
equivalence point	The point in a titration at which the added titrant exactly neutralizes the analyte
half-equivalence point	The point at which half the analyte is neutralized; $\text{pH} = \text{pK}_a$
indicator	A weak acid whose color changes over a pH range; used to detect the endpoint
titration curve	A plot of pH versus volume of added titrant

Answer Key

Worked solutions for every question — see answers embedded in Parts A–C above. The full key with step-by-step work is included in the teacher edition.

Part A — Buffer pH Calculations (Questions 1–6)

- $pK_a = 4.74$; $pH = 4.74 + \log(0.200/0.200) = 4.74$
- (a) $pH = 4.74 + \log(0.600/0.200) = 4.74 + 0.48 = 5.22$. (b) $pH > 7$? No — $5.22 < 7$, so acidic (buffer of a weak acid, even with excess conjugate base).
- $pK_a(NH_4^+) = 14 - pK_b = 14 - 4.74 = 9.26$. $pH = 9.26 + \log(0.250/0.350) = 9.26 - 0.15 = 9.11$
- $5.00 = 4.74 + \log([A^-]/0.50) \rightarrow \log([A^-]/0.50) = 0.26 \rightarrow [A^-] = 0.50 \times 10^{0.26} = 0.91 \text{ M}$. Moles = $0.91 \times 0.300 = 0.27 \text{ mol} \rightarrow \text{mass} = 0.27 \times 82.03 = 22 \text{ g NaCH}_3\text{COO}$ (anhydrous).
- $pH = pK_a$ when $[NH_4^+] = [NH_3]$, so $[NH_4^+] = 0.100 \text{ M}$. Moles = $0.100 \times 0.750 = 0.0750 \text{ mol}$; mass = $0.0750 \times 53.49 = 4.0 \text{ g NH}_4\text{Cl}$.
- Initial $pH = pK_a = 3.17$. Added OH^- converts HF to F^- : $[HF] = 0.39 \text{ M}$, $[F^-] = 0.41 \text{ M}$. $pH = 3.17 + \log(0.41/0.39) = 3.17 + 0.02 = 3.19$.

Part B — Salt Hydrolysis (Questions 7–11)

- (a) acidic (Fe^{3+} hydrolyzes) (b) basic (CO_3^{2-} is conjugate base of HCO_3^-/H_2CO_3) (c) acidic (NH_4^+) (d) neutral (K^+ and ClO_4^- both neutral)
- (a) acidic (b) neutral (c) basic (HCO_2^-) (d) acidic ($CH_3NH_3^+$)
- $CH_3COO^-(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$. Produces $OH^- \rightarrow$ basic solution.
- The product salt NH_4Cl contains NH_4^+ , the conjugate acid of weak base NH_3 , which hydrolyzes to produce H_3O^+ : $NH_4^+(aq) + H_2O(l) \rightleftharpoons NH_3(aq) + H_3O^+(aq)$. The excess H_3O^+ makes the solution weakly acidic.
- The product salt $NaCH_3COO$ contains CH_3COO^- , the conjugate base of weak acid CH_3COOH , which hydrolyzes to produce OH^- : $CH_3COO^-(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + OH^-(aq)$. The excess OH^- makes the solution weakly basic.

Part C — Titrations (Questions 12–16)

- Find the pH at the equivalence point (below 7 for a weak base + strong acid), then choose an indicator whose color-change range brackets that pH (e.g., methyl red, pH 4.4–6.2, for a weak base titrated with HCl).
- An indicator is a weak acid HIn whose color depends on the ratio $[In^-]/[HIn]$. That ratio changes gradually with pH (Henderson–Hasselbalch), so the visible color shifts over roughly $pK_a \pm 1$ pH units.
- Half-equivalence point: $[CH_3COOH] = [CH_3COO^-] \rightarrow pH = pK_a = 4.74$.
- Equivalence point: total volume 80 mL, $[CH_3COO^-] = 0.0040 \text{ mol} / 0.080 \text{ L} = 0.050 \text{ M}$. $K_b = 5.6 \times 10^{-10}$; $[OH^-] = \sqrt{(5.6 \times 10^{-10} \times 0.050)} = 5.3 \times 10^{-6} \text{ M}$; $pOH = 5.28$; $pH = 8.72$.
- Excess $OH^- = 0.001 \text{ L} \times 0.100 \text{ M} = 1.0 \times 10^{-4} \text{ mol}$ in $0.081 \text{ L} \rightarrow [OH^-] = 1.2 \times 10^{-3} \text{ M}$; $pOH = 2.91$; $pH = 11.09$.